Stage 1: Reference Anchoring

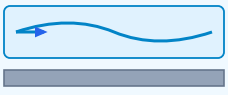
(once per cooling primitive)

Reference Data:
CFD, Vendor
Specification, Experiment

→

Anchor Cooling Primitives
(h_cp, h_eff, u_D)

Cooling Primitives (D2C plate, μCH+Cu-TGV, Cu-pillar pin-fin)



D2C plate



μCH + Cu-TGV

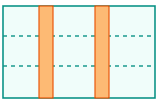


Cu-pillar pin-fin

Stage 2: Design Exploration

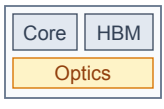
(a) Inputs

Glass Material Library



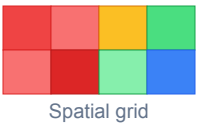
k_glass(T)
TPV vias
Cavities

Floorplans



Multi-die
stack map

Trace-derived power maps



Spatial grid



Dynamic trace P(t)

- Per-core active power
- Transient burst profiles

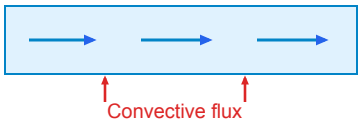
(b) Compact Thermal Engine (3D-ICE 4.0 / 3D-ICE-Glass)

RC Network



Non-uniform cell mesh
Conductive/capacitive

4RM coolant



4-Resistor model
1D/2D laminar flow

OpenMP / SuperLU MT solver

Sparse matrix factorization for fast steady-state & transient sweeps

(c) Outputs

Component Peaks and Feasibility *(T_max < T_lim,i)*



Thermal Headroom H
*and limiting block i**



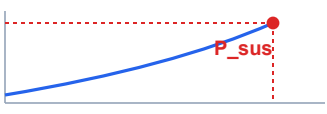
Min headroom at block i*

Optical-region spread
ΔT_lat



PIC lateral gradient

Static capacity *P_sus*



Max steady-state capacity